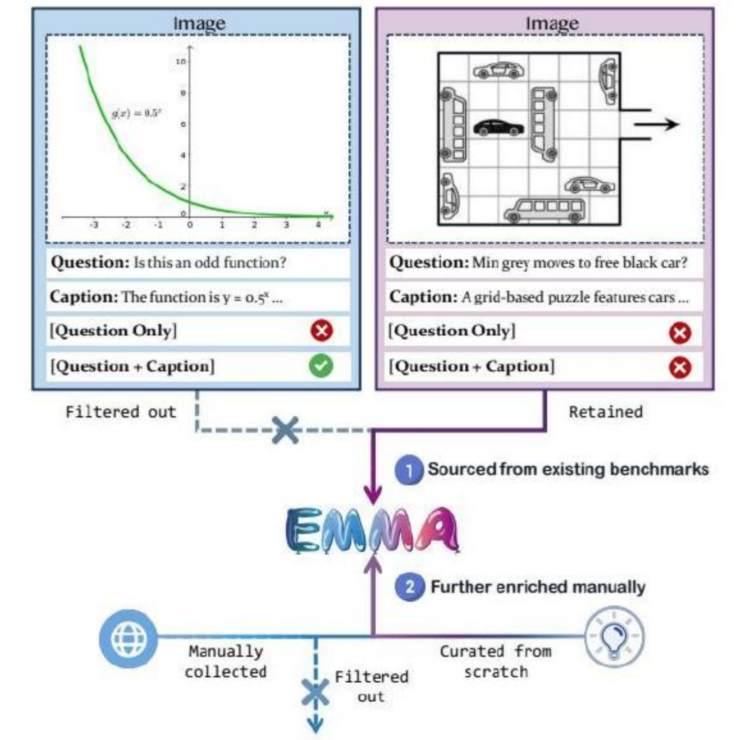
Can MLLMs Reason in Multimodality? EMMA Benchmark

UESTC; SYSU; UW; Microsoft; CUHK

Problem Overview

Setting and Motivation

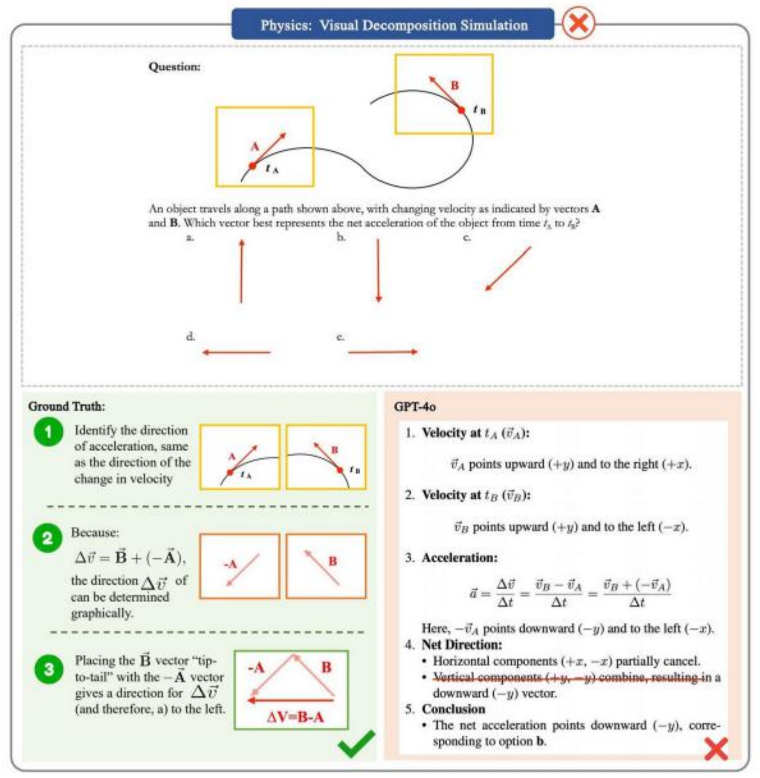


Force Direction

Fig. 1: Force direction

Curved path with v arrows: where is net acceleration?

Read diagram; relate curvature to centripetal a and dv



Key Definitions

Scope: math, physics, chem, coding; MCQ + free

Organic multimodal: vision + multi-step reasoning

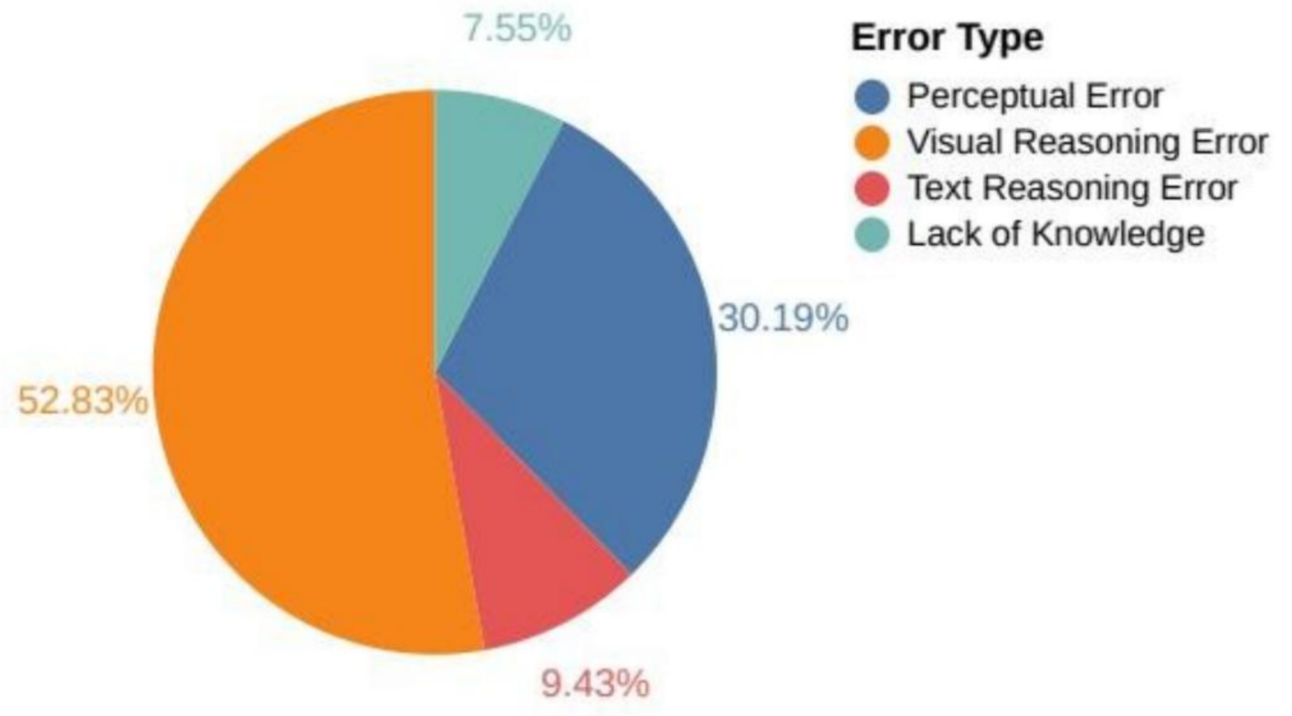
Objective: accuracy vs. human; per-skill diagnostics

Why Methods Fail

Visual Reasoning Dominates

Over half of errors visual: spatial and transformations

CoT helps at times, scaling or tuning won't integrate



Prompts & Compute

Test-time scaling: sample N=1,2,4,8,16; rerank/select

Standard splits + unified prompting: MCQ+open; dir/CoT

Gains plateau; big gap on spatial and multi-hop tasks

Curation & Labels

Subject-specific data + checks; plausible distractors

Two-step curation: captions + rules purge shortcuts

Balanced stats; pressures visual-to-symbolic reasoning

EMMA’s Question: Can MLLMs Reason Multimodally?

UESTC; Sun Yat-sen Univ.; UW; Microsoft; CUHK

EMMA Metrics Gap

Metrics and Human Gap

EMMA/mini: overall, per-category accuracy; baselines

Path, 3D, multi-hop, pattern: large human gap

Identical prompts; strong distractors expose reasoning

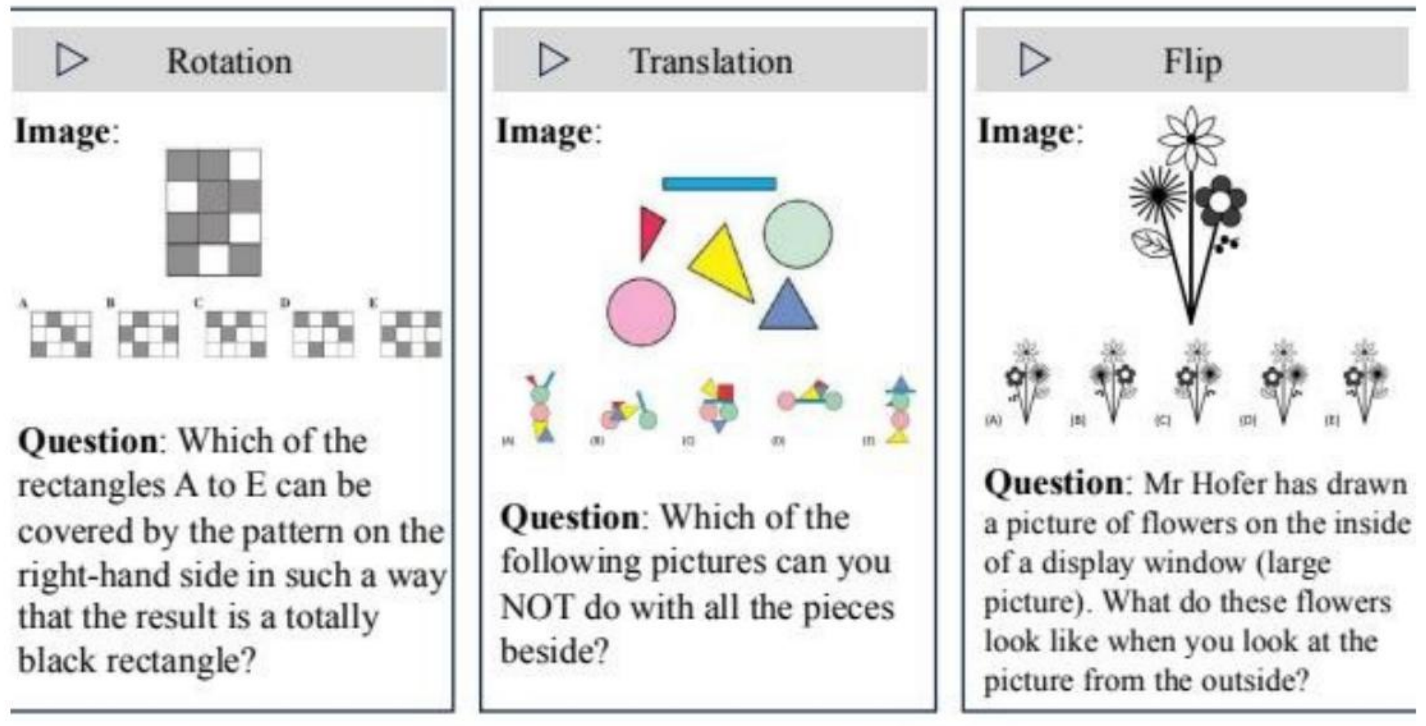
	EMMA-mini	EMMA					
	Overall (100)	Path (13)	3D (37)	MH (33)	VD (47)	GR (26)	Overall (156)
Random choice	23.00	38.46	21.62	27.27	31.91	11.54	25.64
Human Expert	64.50	-	-	-	-	-	-
Direct Claude 3.5 Sonnet	34.00	30.77	37.84	36.36	31.91	30.77	33.97
Direct Gemini 2.0 Flash	40.00	38.46	29.73	42.42	38.30	46.15	38.46
Direct GPT-4o	38.00	30.77	40.54	33.33	36.17	50.00	38.46
Direct Qwen2-VL-72B-Instruct	40.00	30.77	48.65	45.45	42.55	34.62	42.31
Direct LLaVA-Onevision-72B	32.00	23.08	27.03	39.39	48.94	26.92	35.90
Direct InternVL2-Llama3-76B	22.00	15.38	32.43	21.21	6.45	23.08	22.44
Direct InternVL2.5-78B	40.00	38.46	43.24	42.42	38.30	26.92	38.46
CoT Claude 3.5 Sonnet	38.00	53.85	37.84	36.36	42.55	42.31	41.03
CoT Gemini 2.0 Flash	41.00	30.77	29.73	36.36	44.68	46.15	38.46
CoT GPT-4o	44.00	69.23	35.14	39.39	44.68	46.15	43.59
CoT Gemini 2.0 Flash Thinking-1219	57.00	61.54	43.24	57.58	61.70	61.54	56.41
CoT Gemini 2.0 Flash Thinking-0121	63.00	61.54	54.05	60.61	57.45	73.08	60.26
CoT OpenAI o1	49.00	-	-	-	-	-	-
CoT Qwen2-VL-72B-Instruct	34.00	15.38	35.14	33.33	34.04	46.15	34.62
CoT LLaVA-Onevision-72B	26.00	15.38	10.81	18.18	6.38	15.38	12.18
CoT InternVL2-Llama3-76B	33.00	53.85	27.03	21.21	12.90	38.46	32.05
CoT InternVL2.5-78B	36.00	46.15	35.14	30.30	46.81	42.31	39.74

Filtering Rules

Enforce Visual Reasoning

Not solvable from captions; plausible visual foils

Multi-step: rotation, symmetry; minimal domain facts



Negative Examples

Wedge rotations: miss global rule; chase local cues

Cube net: opposite faces/sums; many choose wrong folds

Tiny cases need precise, multi-step visual integration

Results Summary

Gaps and Gains at a Glance

MLLMs trail humans on EMMA and EMMA-mini

Largest gaps: 3D/spatial sim, planning, multi-hop

Test-time scaling helps; the human gap persists

One-pass prompting strong; fails on visual relations

Bottleneck: visual reasoning, not perception

	EMMA-mini	EMMA					
	Overall (100)	SR (474)	GR (9)	RS (132)	RS-pro (105)	KS (456)	Overall (1176)
Random choice	27.00	24.47	33.33	27.27	35.24	0.44	16.50
Human Expert	86.00	-	-	-	-	-	-
Direct Claude 3.5 Sonnet	44.00	66.88	22.22	55.30	55.24	6.80	40.90
Direct Gemini 2.0 Flash	36.00	54.01	11.11	53.79	58.10	8.33	36.31
Direct GPT-4o	33.00	47.05	11.11	51.52	42.86	8.33	31.89
Direct Qwen2-VL-72B-Instruct	34.00	45.99	33.33	48.48	45.71	9.65	32.06
Direct LLaVA-Onevision-72B	24.00	38.19	33.33	39.39	26.67	7.24	25.26
Direct InternVL2-Llama3-76B	21.00	37.34	22.22	31.45	24.76	8.55	24.06
Direct InternVL2.5-78B	38.00	55.06	33.33	47.73	43.81	8.99	35.20
CoT Claude 3.5 Sonnet	41.00	57.17	33.33	58.33	58.10	15.57	41.07
CoT Gemini 2.0 Flash	36.00	22.15	33.33	50.00	59.05	11.84	24.66
CoT GPT-4o	35.00	42.41	33.33	51.52	45.71	16.67	33.67
CoT Gemini 2.0 Flash Thinking-1219	41.00	48.31	33.33	45.45	69.52	17.76	37.93
CoT Gemini 2.0 Flash Thinking-0121	47.00	53.16	66.67	48.48	58.10	23.25	41.58
CoT OpenAI o1	40.00	-	-	-	-	-	-
CoT Qwen2-VL-72B-Instruct	32.00	33.33	11.11	37.12	42.86	7.89	24.57
CoT LLaVA-Onevision-72B	23.00	33.76	0.00	37.88	20.95	7.24	22.53
CoT InternVL2-Llama3-76B	21.00	29.11	22.22	30.65	22.86	6.58	19.73
CoT InternVL2.5-78B	24.00	37.13	33.33	37.12	33.33	13.16	27.47